

Preliminary Results:

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For this part of the course project, we have created 2 dense neural networks using our dataset which we used to predict our labels. We will discuss the aspects of the neural network, evaluate them using learning curves on training and validation sets, and potentially suggest and explore feature engineering steps that we will implement in the final project.

Aspects of our neural networks;

Model 1. Neural Network Specifications and Choices:

- **Number of Layers:**
 - Model 1: 3 layers (input, hidden, and output).

```
model <- keras_model_sequential()  
model %>%  
  layer_dense(units = 50, activation = "relu", input_shape =  
ncol(training_features)) %>%  
  layer_dense(units = 25, activation = "relu") %>%  
  layer_dense(units = 1, activation = "sigmoid")
```

- Model 2: 3 layers but with more units in the hidden layers.

```
model2 <- keras_model_sequential()  
model2 %>%  
  layer_dense(units = 100, activation = "relu", input_shape =  
ncol(training_features)) %>%  
  layer_dense(units = 50, activation = "relu") %>%  
  layer_dense(units = 1, activation = "sigmoid")
```

- The difference between the two models are the first hidden layer, 50 vs 100 and the second hidden layer of 25 vs 50.
- The choice of layers aims to balance model complexity and computational efficiency. More layers and units can capture complex patterns but risk overfitting.
- **Number of Units per Layer:**
 - Model 1: 50 units in the first hidden layer and 25 in the second.
 - Model 2: 100 units in the first hidden layer and 50 in the second.
 - These were chosen to provide sufficient capacity to learn from the data while controlling overfitting through fewer units.

- Model 1 has fewer parameters and is less likely to have over fitting but it also might underfit due to having less layers thus the creation of model 2.

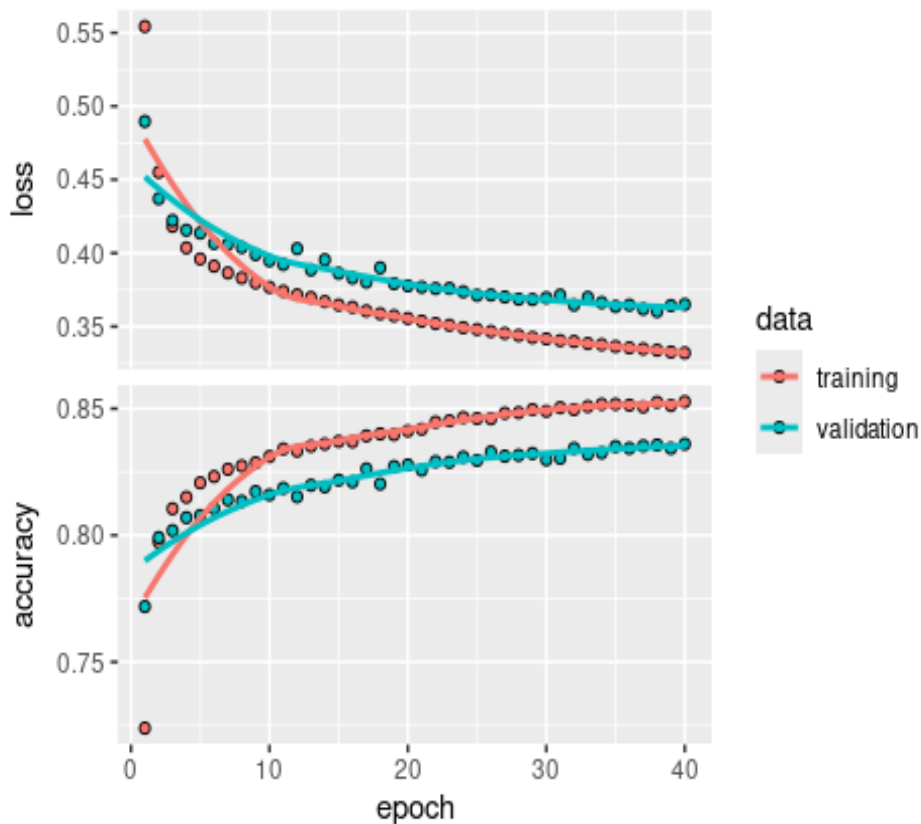
- **Activation Functions:**

- Relu (Rectified Linear Unit) for our hidden layers: This is a very common choice for hidden layers as it introduces non-linearity to our model..
- Sigmoid for the output layer: Appropriate for binary classification problems to produce probability scores which we will be able to use to evaluate our models

- **Loss Function:**

- Binary Crossentropy was used as it is standard for binary classification tasks, measuring the difference between true labels and predicted probabilities.

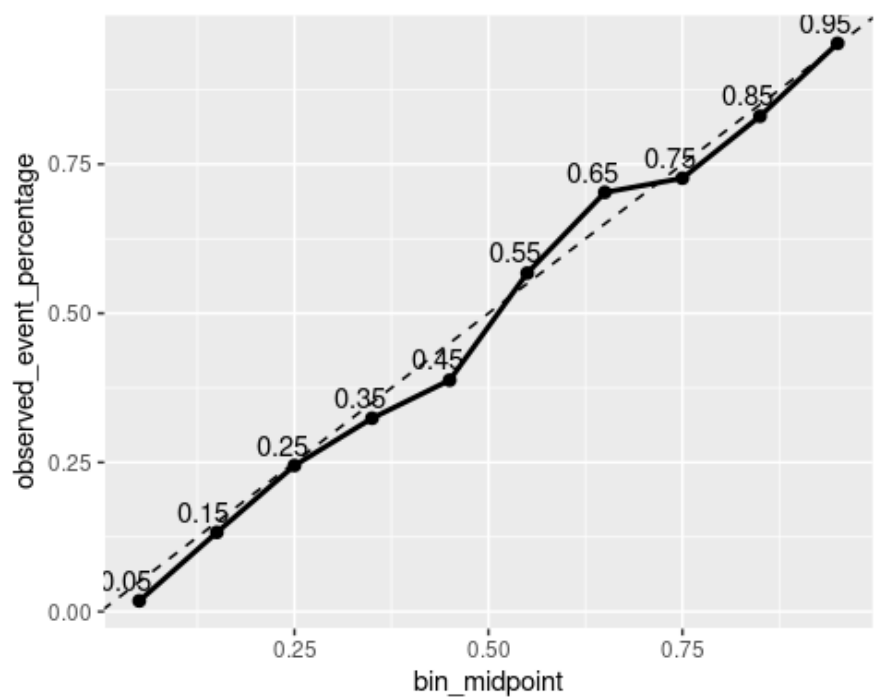
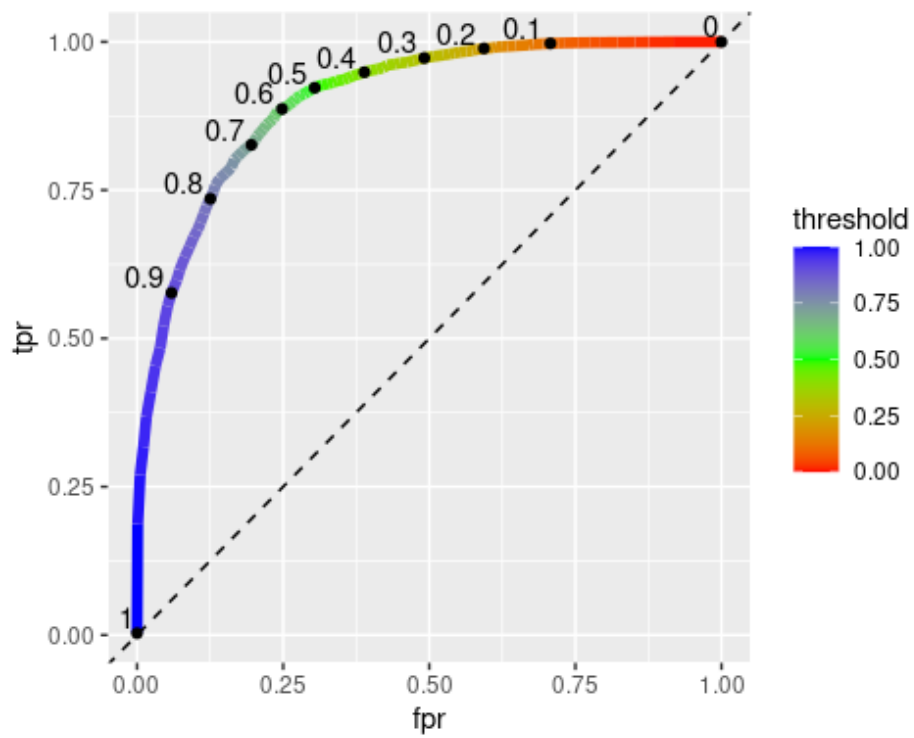
- **Optimization Algorithm:**



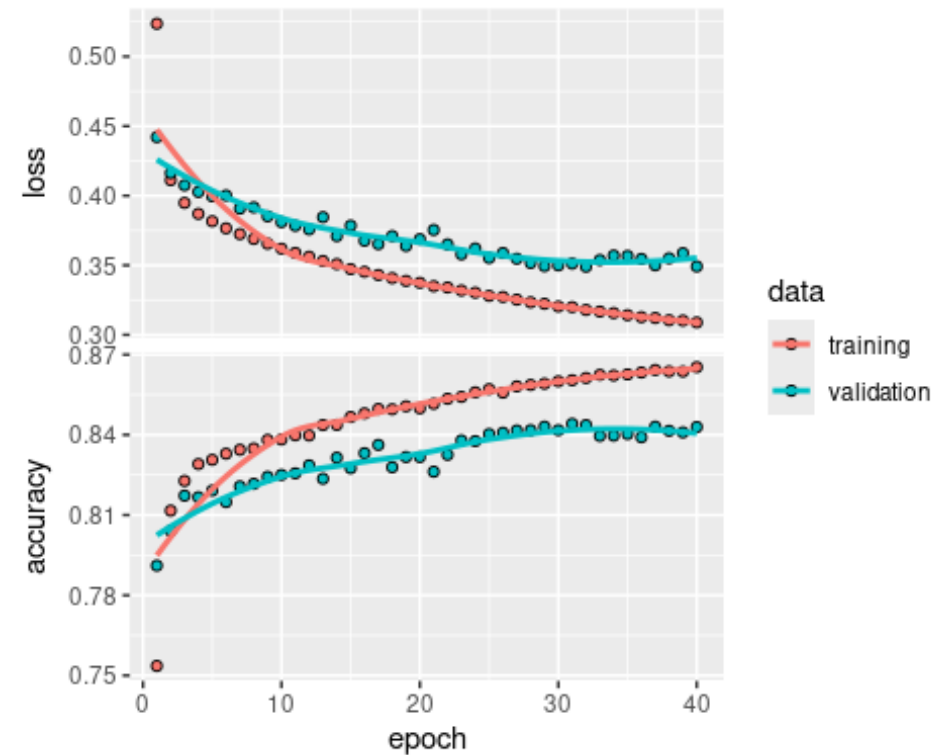
- RMSprop was selected for its effectiveness in dealing with noisy gradients and maintaining a steady learning rate

Model 1:

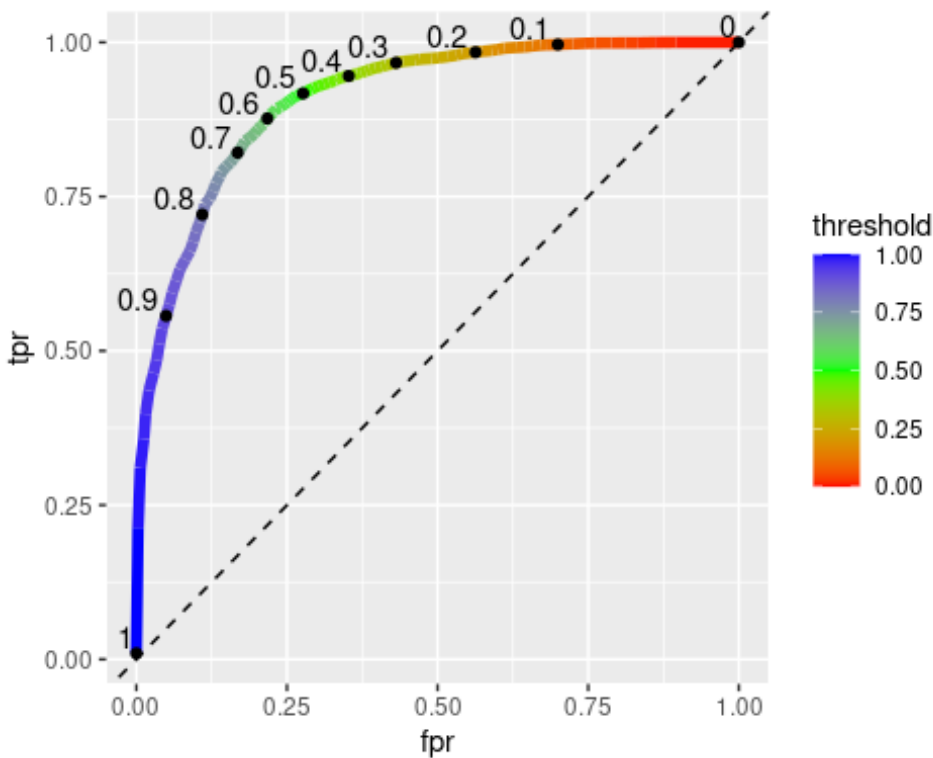
When we plotted our loss and accuracy curves for both training and validation sets we can observe that training loss is lower than the validation loss, it indicates overfitting. But this is not too significant to raise any red flags.

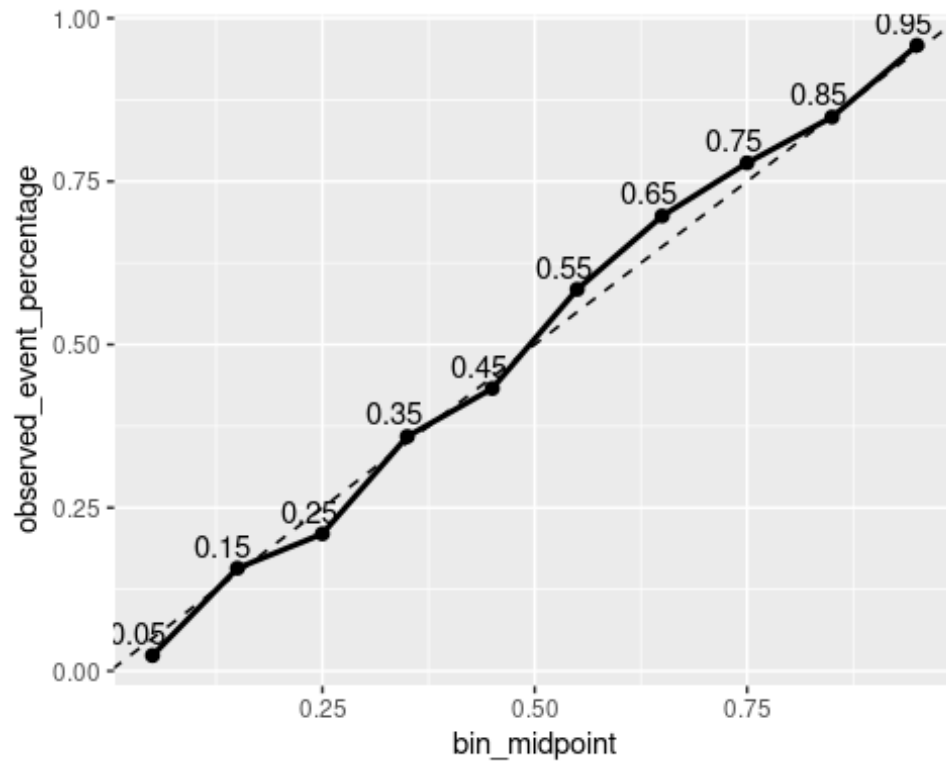


Model 2:



When we plotted our loss and accuracy curves for both training and validation sets we can observe that training loss is lower than the validation loss, it indicates overfitting. But this is not too significant too raise any red flags.





Outputs of our model:

Model 1

- **AUC:** 0.903724
- **False Positive Rate (FPR) at 0.5 threshold:** 0.3039573
- **True Positive Rate (TPR) at 0.5 threshold:** 0.9225506
- **Test Set Accuracy:** 0.7379532
- **Training accuracy:** ~81.6%
- **Validation accuracy:** ~ 81.8%
- **Validation loss:** ~ 0.38
- **Training loss:** ~ 0.38

Model 2

- **AUC:** 0.9105204
- **False Positive Rate (FPR) at 0.5 threshold**
- **True Positive Rate (TPR) at 0.5 threshold:**
- **Test Set Accuracy:** 0.7384121
- **Validation accuracy:** ~82.4%
- **Traning accuracy:** ~82.3%
- **Validation loss:** ~ 0.37
- **Training loss:** ~0.37

Analysis

- AUC: Model 2 has a slightly higher AUC (0.9105204) compared to Model 1 (0.903724), indicating better performance in distinguishing between classes.
- Test Set Accuracy: Model 2 has a marginally better test set accuracy compared to Model 1.
- Training and Validation Losses: Model 2 shows slight improvements in both training and validation accuracy and has slightly lower training and validation losses compared to Model 1.

Feature engineering for final:

- **Model 2 will be used for the final project but more epochs will be used and i will also build another model with even more layers just to compare what happens. If overfitting occurs and the model performs poorly for the AUC, FPR and TPR then we will stick to model 1.**
- **Booking status_cancelled and not cancelled were created and i would like to test the model by using one individually as a predictor.**

Conclusion

Model 2 is better overall based on AUC, training and validation metrics, and test set accuracy. There are no strong indications of overfitting or underfitting as the validation metrics are close to the training metrics.